# Switch Order Automation — Technical Instructions

**Prepared for:** Automation Intern
**Purpose:** Automate the generation of fund switching orders for HVP portfolio rebalancing

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## Overview

When the portfolio manager updates fund allocations (e.g. removing Fund A and adding Fund B), existing client holdings need to be **switched** from the old fund to the new fund. This script automates the calculation of how many units each client switches out of each fund, and into which fund(s), based on the difference between the old and new allocation files.

The output is a single Excel file containing the switch trades ready for upload by the operations team, plus a manager check tab.

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## Inputs (4 files, prompted in order)

### Input 1 — Rebalance Input File (`.xlsm`)

**What it is:** A snapshot of all client holdings before rebalancing.
**One sheet only.** Each row = one fund holding for one account.

| Column | Description |
|---|---|
| `code` | 6-digit combined fund code (first 3 = MgrCd, last 3 = FundCd) e.g. `617318` |
| `svc_acct` | Client account number |
| `port_type` | Portfolio type e.g. `HVP3`, `HVP3_SRS`, `CBP_SRS` |
| `qty` | Existing units held in this fund |
| `mkt_val` | Market value of holding in SGD |
| `total_pf` | Total portfolio value of this account in SGD (AUM) |
| `ccy` | Fund currency |

**Note:** `FdSrc` is derived from `port_type` — if `port_type` ends with `_SRS`, FdSrc = `SRS-IA`; otherwise `Cash`.

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### Input 2 — Switch Pairing Mapping File (`.xlsx`)

**What it is:** A manually prepared file that specifies which fund switches into which. This is filled in by the portfolio manager based on their judgment of how to match switch-out funds to switch-in funds.

**One sheet, headers as follows:**

| Column | Description | Example |
| --- | --- | --- |
| `port_type` | Portfolio type | `HVP3` |
| `MgrCd_OUT` | Manager code of fund being switched OUT | `618` |
| `FundCd_OUT` | Fund code of fund being switched OUT | `175` |
| `MgrCd_IN` | Manager code of fund being switched INTO | `618` |
| `FundCd_IN` | Fund code of fund being switched INTO | `176` |

**Rules:**

- One row per (source → destination) pair per portfolio type
- If a source fund splits across multiple destinations, add one row per destination
- If a source fund maps to only one destination, it gets one row and 100% of units go there
- If a source fund maps to multiple destinations, units are split proportionally by the destination fund's weight delta (new_weight − old_weight)

**Example mapping for HVP3:**

| port_type | MgrCd_OUT | FundCd_OUT | MgrCd_IN | FundCd_IN |
| --- | --- | --- | --- | --- |
| HVP3 | 618 | 175 | 618 | 176 |
| HVP3 | 617 | 318 | 618 | 167 |
| HVP3 | 617 | 791 | 618 | 167 |
| HVP3_SRS | 618 | 175 | 618 | 167 |
| HVP3_SRS | 617 | 312 | 618 | 167 |
| HVP3_SRS | 617 | 312 | 618 | 176 |
| HVP3_SRS | 617 | 791 | 618 | 167 |

### Input 3 — New Portfolio Allocation File (`.xlsx`)

**What it is:** The updated target allocation — what the portfolio should look like AFTER rebalancing.

Each portfolio type is a separate sheet (e.g. `HVP3`, `HVP3_SRS`, `CBP_SRS`).
Each sheet has these columns:

| Column | Description |
|---|---|
| `mgr_code` | Manager code (3 digits) |
| `fund_code` | Fund code (3 digits, zero-padded e.g. `012` not `12`) |
| `weights` | Target weight as decimal (e.g. `0.35`) OR as % integer (e.g. `35`) — script handles both |
| `fund name` | Full fund name |
| `ccy` | Fund currency |

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### Input 4 — Old Portfolio Allocation File (`.xlsx`)

**What it is:** The previous target allocation — what the portfolio looked like BEFORE rebalancing. Same structure as Input 3.

Used to calculate the delta for each fund:

```
delta = new_weight - old_weight
```

- **delta < 0** → fund is being reduced → switch OUT
- **delta > 0** → fund is being added/increased → switch IN
- **delta = 0** → no change → ignore

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## Core Calculation Logic

### Step 1 — Calculate weight delta per fund per portfolio type

For each portfolio type, compare old and new allocation files:

```
delta = new_weight - old_weight
```

Funds with **delta** < **0** are switch-out funds.
Funds with **delta** > **0** are switch-in funds.

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### Step 2 — Calculate units switched out per account per source fund

For each account, for each switch-out fund the account holds:

```
# Partial switch out (fund still exists in new allocation at lower weight)
units_out = (old_weight - new_weight) / old_weight × existing_qty

# Full switch out (fund removed entirely from new allocation, new_weight = 0)
units_out = existing_qty   ← use full qty to avoid leaving fractional units
```

**Example (HVP3, account 1163485):**

- 617/318 Capital Group Z: old=30%, new=20% → units_out = (30−20)/30 × 110.133 = **36.711 units**
- 617/791 JPM China: old=5%, new=0% → units_out = full qty = **31.428 units**
- 618/175 PineBridge: old=10%, new=0% → units_out = full qty = **100.859 units**

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### Step 3 — Split units across switch-in destinations using the mapping file

For each source fund, look up its destination(s) in the mapping file.

**If one destination:** all `units_out` go there.

**If multiple destinations:** split proportionally by destination fund's delta:

```
units_to_fund_B = units_out × (delta_B / sum_of_deltas_for_this_source)
```

Last destination gets the remainder to avoid rounding residuals.

**Example (HVP3_SRS, 617/312 splits to two destinations):**

- 618/167 delta = +15%, 618/176 delta = +10%, total = 25%
- → 618/167 gets 15/25 = 60% of units
- → 618/176 gets remainder = 40% of units

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### Step 4 — Generate one trade row per (account × source fund × destination fund)

This is a many-to-one or many-to-many depending on mapping. Each row in the Trades tab represents one switch instruction.

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## Output — Single Excel file: `SwitchingOrder_YYYYMMDD.xlsx`

### Sheet 1: Trades

Clean upload-ready file for operations team. **Values only, no formulas.**

| Column | Description |
|---|---|
| `AccountNo` | Client account number (7 digits, zero-padded) |
| `OrderType` | Always `ESO` |
| `MgrCd` | Manager code of fund being switched OUT |
| `FundCd` | Fund code of fund being switched OUT |
| `FdSrc` | `Cash` or `SRS-IA` (derived from port_type) |
| `SIMgrCd` | Manager code of fund being switched INTO |
| `SIFundCd` | Fund code of fund being switched INTO |
| `NetSalesChg` | Always 0 |
| `SwChg` | Always 0 |
| `DivOpt` | `Reinvest` (or `Withdraw` for MD portfolio types) |
| `Units` | Units to switch out of the source fund for this destination |
| `FACode` | Leave blank |
| `TransDate` | Today's date in YYYYMMDD format |

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### Sheet 2: Check

Same columns A–M as Trades, plus the following manager verification columns:

| Col | Header | Formula/Source |
|---|---|---|
| N | Switch Out Fund Name | Fund name lookup from allocation file using MgrCd+FundCd |
| O | Switch In Fund Name | Fund name lookup from allocation file using SIMgrCd+SIFundCd |
| P | Units before Rebalance (Switch Out Fund) | `qty` from Rebalance Input for this account+fund |
| Q | Total Units Switched Out | Sum of Units across ALL rows with same AccountNo+MgrCd+FundCd |
| R | Units Remaining after Rebalance | `P - Q` |
| S | MV of Fund before Rebalance (SGD) | `mkt_val` from Rebalance Input |
| T | Account Portfolio Amount (SGD) | `total_pf` from Rebalance Input |
| U | Total Switch Out % | `Q / P` (shown as %) |
| V | Switch Out Fund Weight (%) after Rebalance | `((P-Q)/P × S) / T` (shown as %) |
| W | Target Allocation of Switch Out Fund (%) | New weight from new allocation file (shown as %) |
| X | Portfolio Type | `port_type` from Rebalance Input |

**Conditional formatting:**

- V ≈ W (within 0.5%) → green ✓
- V differs from W by more than 0.5% → yellow (review needed)

**Note on V vs W:** V will not exactly equal W because V uses market value (which reflects historical price movements) while W is a weight-based target. A small deviation is expected and normal. Large deviations should be investigated.

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### Sheet 3: New Portfolio Allocation >>

Empty placeholder tab. Portfolio manager fills this in manually if needed.

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### Sheets 4+: Portfolio Allocation Tabs

One tab per portfolio type that appears in the trades (e.g. `HVP3`, `HVP3_SRS`). Copied from the new allocation file with:

- `weights` column converted to % format (e.g. 0.35 → 35%)
- `fund_code` zero-padded to 3 digits (e.g. 12 → `012`)

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## Key Rules and Edge Cases

| Scenario | Rule |
| --- | --- |
| Fund fully removed (new_weight = 0) | Use full `qty` — do not calculate ratio |
| Fund partially reduced | Use `(old-new)/old × qty` |
| Account has no holding in a switch-out fund | Skip — do not generate a row |
| `port_type` ends with `_SRS` | FdSrc = `SRS-IA` |
| `port_type` starts with `MD` | DivOpt = `Withdraw` |
| All other port_types | FdSrc = `Cash`, DivOpt = `Reinvest` |
| Weights in allocation file as % integers (e.g. 35) | Divide by 100 before delta calculation |
| Weights in allocation file as decimals (e.g. 0.35) | Use as-is |
| fund_code column named `weights (%)` | Treat same as `weights` |
| Last destination in a multi-destination split | Gets the remainder to avoid rounding residuals |

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## Script Prompts (in order)

1. Select Rebalance Input File
2. Select Switch Pairing Mapping File
3. Select New Allocation File
4. Select Old Allocation File
5. Select Output Folder

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## What the Script Does NOT Do

- Does not determine which funds to switch — that is set by the old vs new allocation files

- Does not decide which source fund maps to which destination fund — that is in the mapping file, filled by the portfolio manager
- Does not handle buy orders or sell orders — only switches (ESO)
- Does not apply a cash buffer (no FEES multiplier) — switches are purely unit-based on existing holdings